FarmGPU MLPerf Storage v2.0 IntelSPR Single-Host System Description

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System Name: FarmGPU-IntelSPR-Storage01

Submitter Name: FarmGPU

Submission Date: July 11, 2025

Version of the Benchmark: MLPerf Storage v2.0

Solution Type of the Submission: Local Storage

Submission Division: CLOSED

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1. System Topology

The system under test consists of a single host acting as both the DLIO client and the storage data node, connected to a backend switch. This integrated setup simplifies the network architecture for single-host performance testing.

2. Power Requirements

- **DLIO Client & Storage Data Node (Single-Host Node):**
  - Quantity: 1
  - Design Power: 2000W
  - Number of Active PSUs: 2
  - Number of Passive PSUs: 0

Component	Power Supply Unit	Nameplate Power	Design Power
DLIO Client & Storage Data Node	PSU 1	2000 W	2000 W
	PSU 2	2000 W	



3. Hardware

- **3.1 Single-Host Node**
  - **Server Model:** SSG-121E-NES24R Product Page
  - **Form Factor:** 1U Rack Mount
  - **Motherboard:** X13DSF-A
  - **CPU:** 2 x Intel(R) Xeon(R) Platinum 8450H:
    - Total Cores: 28
    - Total Threads: 56 (enabled by Hyper-Threading)
    - Processor Base Frequency: 2.00 GHz
    - Max Turbo Frequency: 3.5 GHz
    - All Core Turbo Frequency: 2.6 GHz
    - Cache: 75 MB (L3)
    - TDP (Thermal Design Power): 250 W
    - Socket Type: FCLGA4677
    - Memory Support: DDR5 (rated speed 4800 MT/s)
    - Memory Channels: 8
    - Maximum Memory Size: 4 TB

        - PCI Express Revision: 5.0
        - Maximum Number of PCI Express Lanes: 80
        - Launch Date: Q1'23 (January 2023)
        - Architecture: Sapphire Rapids-SP
        - Scalability: Up to 8 CPUs in an SMP configuration
    - **Memory:** 256GB (16x16GB).
    - **Storage:** 24 x FADU Echo Gen5 E1.S SSD (FSNE50E1SD7T6W000) 7.68TB PCIe 5.0 NVMe SSDs in a RAID-0 configuration
    - **Networking:**
        - Management: 2 x Intel Corporation Ethernet Controller X710 10Gbps.
        - High-Speed: 2 x Mellanox Technologies MT2910 Family [ConnectX-7].

4. Software

- **4.1 Single-Host Node**
    - **Operating System:** Ubuntu 24.04.2 LTS (Noble Numbat), Kernel Linux 6.15.4 (x86_64)
    - **MPI:** Open MPI 4.1.4

5. Settings

- 5.1 XFS Configuration
  The XFS filesystem was configured with the following parameters:
    - meta-data=/dev/md127 isize=512 agcount=168, agsize=268434944 blks
    - sectsz=4096 attr=2, projid32bit=1
    - crc=1 finobt=1, sparse=1, rmapbt=1
    - reflink=1 bigtime=1 inobtcount=1 nrext64=0
    - data = bsize=4096 blocks=45007994880, imaxpct=1
    - sunit=512 swidth=12288 blks
    - naming =version 2 bsize=4096 ascii-ci=0, ftype=1
    - log =internal log bsize=4096 blocks=521728, version=2
    - sectsz=4096 sunit=1 blks, lazy-count=1
    - realtime =none extsz=4096 blocks=0, rtextents=0
- 5.2 RAID Configuration
  A RAID-0 array was created using 24 NVMe devices (/dev/nvme[1-24]n1).
    - md127 : active raid0 nvme25n1[23] nvme24n1[22] nvme23n1[21] nvme22n1[20] nvme21n1[19] nvme20n1[18] nvme19n1[17] nvme18n1[16] nvme17n1[15] nvme16n1[14] nvme15n1[13] nvme14n1[12] nvme13n1[11] nvme12n1[10] nvme11n1[9] nvme10n1[8] nvme9n1[7] nvme8n1[6] nvme7n1[5] nvme6n1[4] nvme5n1[3] nvme4n1[2] nvme3n1[1] nvme1n1[0]
    - 180031979520 blocks super 1.2 2048k chunks

- 5.3 Mount Override
  The RAID array was mounted with specific XFS options for performance:
  - `/dev/md127 on /mnt/ssd type xfs (rw,noatime,nodiratime,attr2,inode64,logbufs=8,logbsize=32k,sunit=4096,swidth=98304,noquota)`
- 5.4 Memory Allocation Tweaks Performed
- 5.5 VM Tweaks
  Virtual machine settings were optimized:
  - SMT (Simultaneous Multi-threading) was disabled: `echo off | sudo tee /sys/devices/system/cpu/smt/control`
  - `vm.page-cluster` was set to 0: `sudo sysctl -w vm.page-cluster=0`
  - `vm.min_free_kbytes` was set to 262144: `sudo sysctl -w vm.min_free_kbytes=262144`
- 5.6 CPU Governor
  The CPU governor was set to performance mode for all CPUs:
  - `for cpu in /sys/devices/system/cpu/cpu[0-9]*; do echo performance | sudo tee "$cpu/cpufreq/scaling_governor"; done`
- 5.7 GRUB Configuration[1]
  The GRUB command line was modified to disable memory initialization for improved performance. The GRUB_CMDLINE_LINUX_DEFAULT line was changed from:
  - GRUB_CMDLINE_LINUX_DEFAULT=""
    to:
  - `GRUB_CMDLINE_LINUX_DEFAULT="init_on_alloc=0 init_on_free=0"`
    After modification, GRUB was updated and the system rebooted:
  - `sudo update-grub`
  - `sudo reboot`

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1. `init_on_alloc=0` is a setting recommended by NVIDIA to improve performance in these workloads [cite: https://docs.nvidia.com/grace-perf-tuning-guide/os-settings.html#init-on-alloc]